

⇒ Weighted Mean (DS)

The weighted mean is the sum of the products of each value and its weight, divided by the sum of weight.

AI models House price Predict (

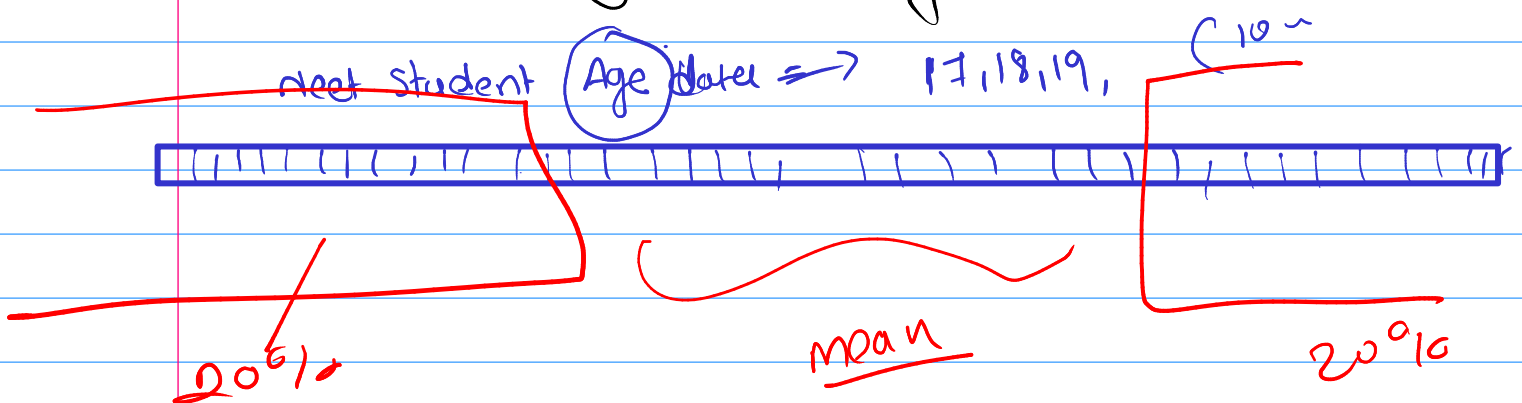
→ LR	→ 0.2% accuracy	= 10L ✓	$\frac{(10 \times 0.2) + (12 \times 0.3) + (15 \times 0.5)}{0.2 + 0.3 + 0.5}$
RF	⇒ 0.3%	⇒ 12L ✓	
Xgboost	⇒ 0.5%	⇒ 15L ✓	

ML-model weight ↓
rise

⇒

⇒ Trimmed Mean

It is calculated by removing a certain percentage of smallest and largest value from the dataset. And then taking the mean of remaining values.



Measure of
Central
tendency

mean

median

mode

Trimmed mean

weight mean.

Naqpus

Measure of Dispersion

spread

- It is a Statistical measure that describe the spread or Variability of a dataset.
- It provide information about how the data is distributed around the central tendency.

① Range $\Rightarrow$

(The Range is difference between the maximum and minimum value in dataset)

- It Robust to outlier.
affected by

$$\text{Age} = 17, 18, 19, 18, 17, 18, 19 \rightarrow \text{Range} = 19 - 17 = \underline{\underline{2}}$$

$$\begin{array}{l} \text{Age} \\ \text{wim} \\ \text{outlier} \end{array} = 17, 18, 19, 18, 17, 18, 19, 19 \rightarrow \text{Range} = \underline{\underline{19 - 17}}$$

② Variance (σ^2) $\rightarrow$ Sigma

The Variance is the average of the Squared difference between each data point and the mean.

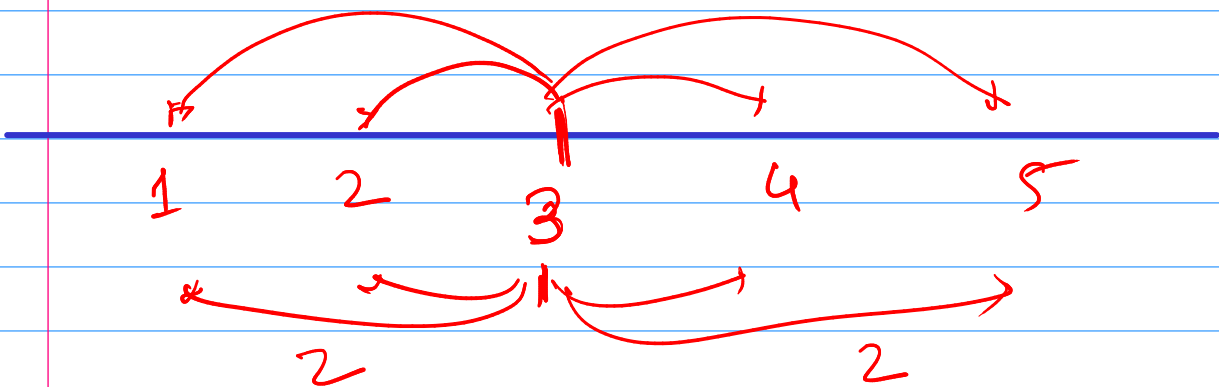
<u>Data Point</u>	$(x - \text{mean})$	$(x - \text{mean})^2$
3	$3 - 3 = 0$	0
2	$2 - 3 = -1$	1
1	$1 - 3 = -2$	4
5	$5 - 3 = 2$	4
4	$4 - 3 = 1$	1

$$\text{mean} = \frac{3+2+1+5+4}{5} = \frac{15}{5} = \boxed{3}$$

$$\frac{0+1+4+4+1}{5}$$

$$= \frac{10}{5} = \boxed{2}$$

Variance = 2



km

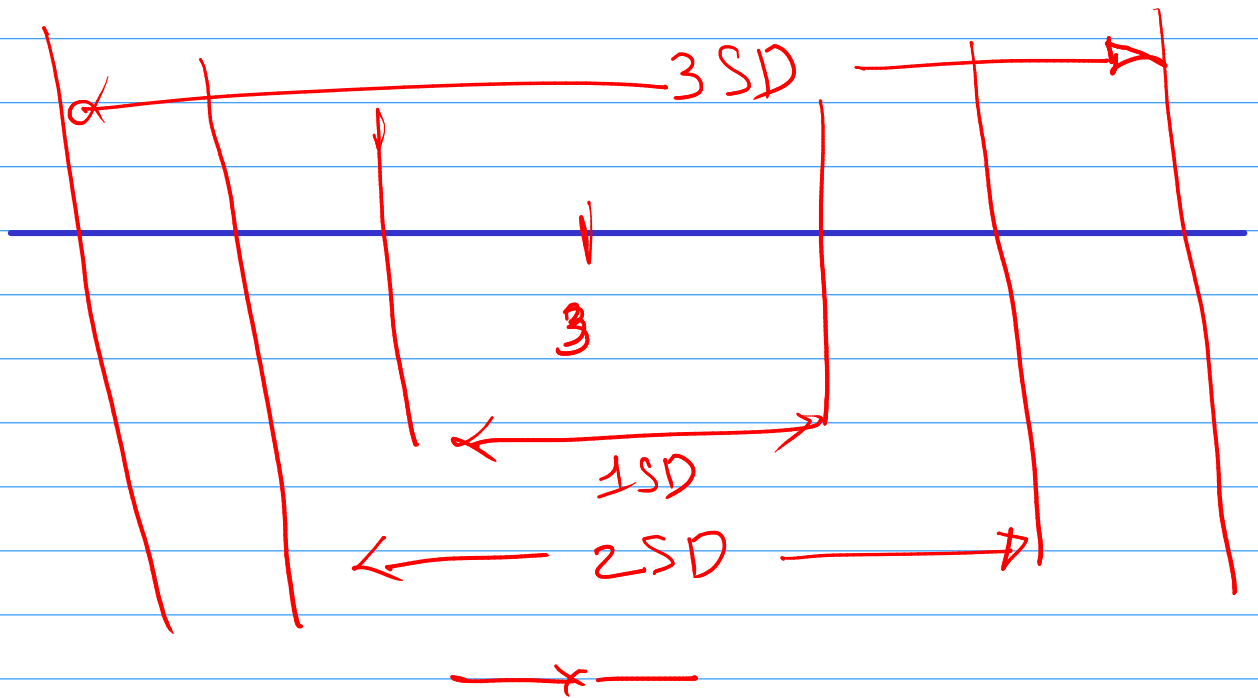
$$\sigma^2 = \underline{\underline{1 \text{ km}^2}} - (\text{unit})^2$$

Unit

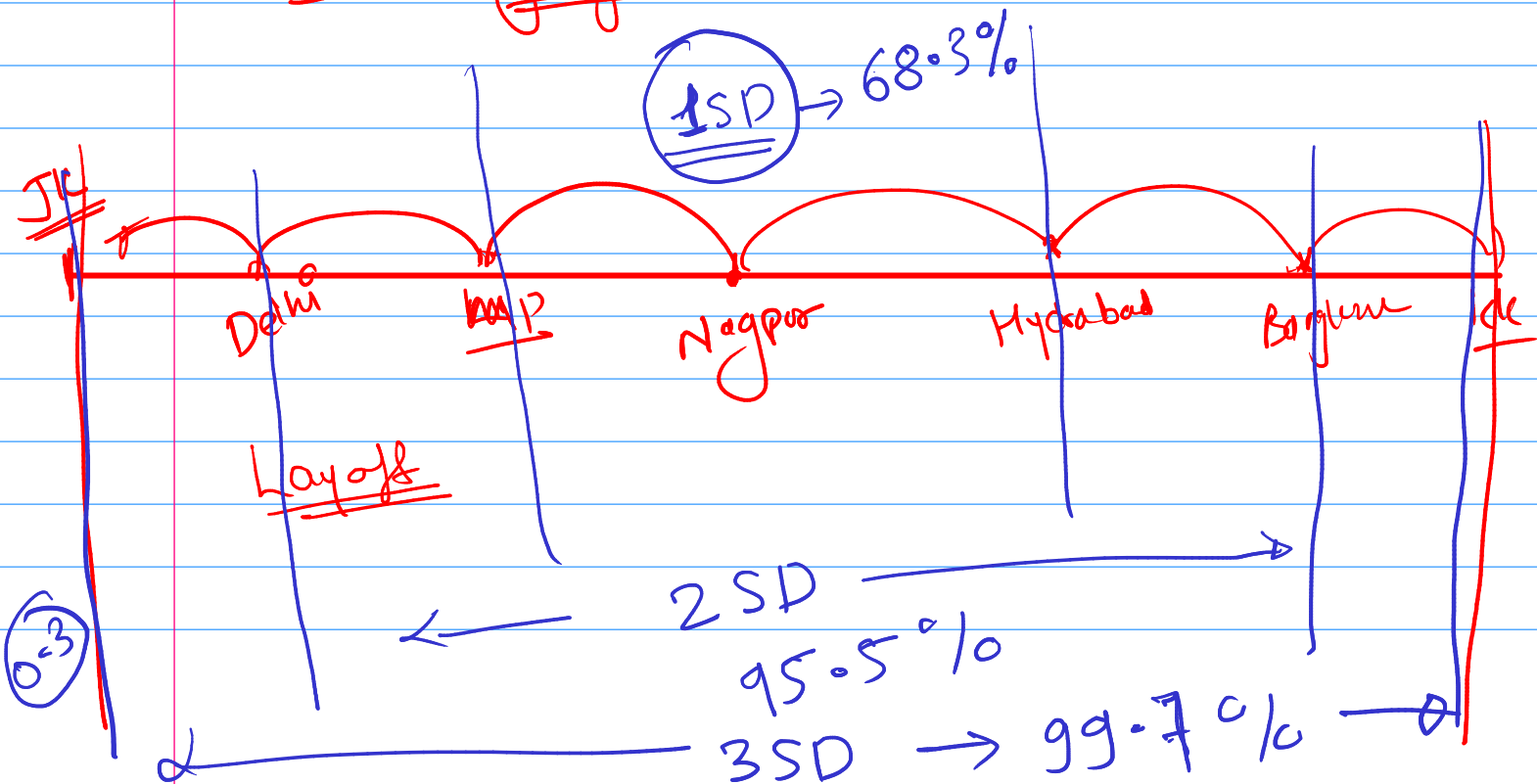
③ Standard Deviation (

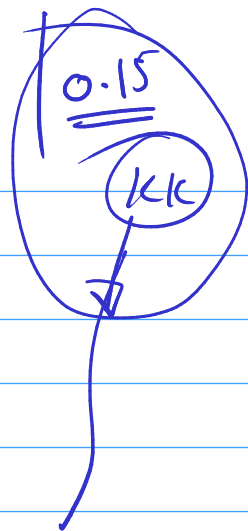
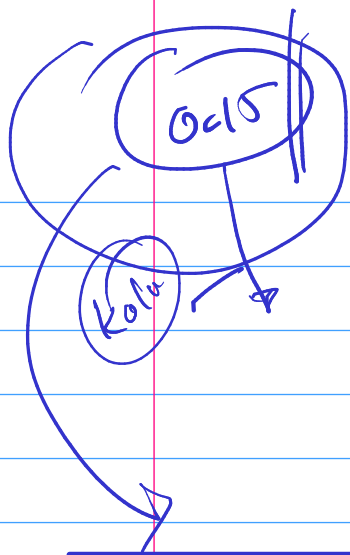
→ One standard deviation is the square root of the variance.

$$SD = \sqrt{\sigma^2}$$



ex → Project





④ Coefficient of Variation (CV)

→ The CV is the ratio of SD and the mean express in percentage.

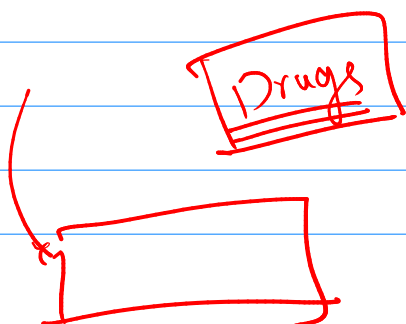
$$CV = \left(\frac{SD}{\text{mean}} \right) \times 100$$

Unit - different

→ Biology, Chemistry & Engg.

Salary (lakh)	Year of exp
1	1
2	2
3	3
4	4

1000 →
(1 year)



(g) (mg)

→ g

10g → 10mg

10mg → 10mg

Sampling Techniques

A. Simple Random Sampling

⇒ we select random samples from the population for study purpose.

ex ⇒ selecting 100 employees randomly from 10000 employees group.

B. Stratified Sampling

⇒ Gender Strategy, Age Strategy —

ex ⇒ Bank loan

→ Car — Bike → Home

Income bracket

C. Systematic Sampling

ex ⇒ Janganand } 1, 2, 3, 4,

→ nirva ⇒ Roll No.

D. Convenience Sampling

↳ Blood Camp :-

↳ Credit Card.

E. Purposive Sampling

→ It involves in Research.

↳ ex → Covid 19

F. Cluster Sampling

Area

↳ Kandha District

↳ exit poll → 100 feed back.

Kandha To.

Selburo

Anu i

Wu

Sun